Centralizers of a maximal block action

Human Compression project: a component of the L044 recovery

Let $G = A_n = \langle \rho_0, \ldots, \rho_{r-1} \rangle$ in its natural action, with the distinguished involutions satisfying the string commutations and intersection property. Put $H = G_i = \langle \rho_j : j \neq i \rangle$. This paper proves the following part of the main rank bound.

Theorem 1. *Suppose $n \geq 12$, H is transitive, and a maximal nontrivial block system for H has $m > 2$ blocks of size $k > 2$. Then*

$$r \leq \lfloor (n-1)/2 \rfloor.$$

The source is Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Proposition 3.2. We acquire the complete maximal-block argument from the reviewed L047 recovery, including Whiston's independent-family bound and its equality clause. We also acquire two precisely bounded finite conclusions: no transitive subgroup of A_6 has a rank-four string C-marking, from the complete L047 proper-group census and its separate A_6 exclusion; and A_{12} has no marking of rank greater than five, from the complete L071 census. These are retained theorems with their proofs and execution records, not new computations here.

The new work below is an equality argument for the block decomposition. In particular, we do not require the general assertion that an omitted mark cannot commute with a subgroup merely because its block image is primitive. Faithfulness and the centralizer are established when needed.

1 The block decomposition and its equality cases

Choose an inclusion-minimal distinguished subfamily L whose images generate the primitive block group P. Let C be the distinguished marks outside L commuting with all of L, and let R be the remaining marks of H. Write $B = \langle L \rangle$ and $D = \langle C \rangle$. The acquired argument has the following exact consequences. If the image diagram of L is disconnected, then $m \geq 60$ and

$$r \leq 2 \log_2 m + k + 4.$$

This is smaller than $(km-1)/2$ for $k \geq 3, m \geq 60$: the difference is increasing in both variables and is already positive at $(3, 60)$. We may therefore suppose that L occupies an interval of labels. In that case

$$|L| \leq m - 1, \qquad |C| \leq k - 1, \qquad |R| \leq 2, \qquad r \leq k + m. \tag{1}$$

The last inequality includes the acquired exclusion of $\operatorname{rank}(H) = k + m$.

The group D fixes each block and has a faithful restriction to every block. Indeed, the primitive group P, generated by involutions, has trivial centralizer in S_m when $m > 2$. A nontrivial primitive centralizer would force P to be regular of prime degree; the cyclic group of odd prime order is not generated by involutions. Thus D fixes the blocks. An element of D fixing one block pointwise fixes every block, by commutation with the block-transitive group B. This proves faithfulness. Transporting the coordinates of one block by elements of B identifies D with the same permutation group on each block.

The intersection property gives $B \cap D = 1$, and every distinguished subfamily remains independent. If $|C| = k - 1$, Whiston's equality clause therefore makes D the diagonal S_k. Its centralizer in the full block stabilizer is S_m acting consistently on the block coordinates. Consequently B is faithful on the blocks and is exactly the diagonal copy of P in these coordinates.

Lemma 2. *If $r = k + m$, then*

$$|L| = m - 1, \qquad |C| = k - 2, \qquad |R| = 2. \tag{2}$$

In particular $P = S_m$ and ρ_i commutes with B.

Proof. Equation (1) and $r - 1 = |L| + |C| + |R|$ leave exactly three possibilities:

$$(|L|, |C|, |R|) = (m - 1, k - 1, 1),\ (m - 2, k - 1, 2),\ (m - 1, k - 2, 2).$$

Consider the second. Here B is a faithful diagonal copy of the primitive group P. Since P has trivial centralizer, the full centralizer of B on the km points consists of permutations of its k identical rows, acting consistently on them. It fixes each of the original blocks. Both neighbours of the interval L are occupied by the two marks in R. Hence the omitted label i is not adjacent to L, so ρ_i centralizes B and preserves the blocks. Then $G = \langle H, \rho_i \rangle$ preserves them, contrary to primitivity.

In the first possibility, $B = S_m$ and $D = S_k$ form their product action. The noncommuting diagram of C is connected: disconnected components would give a nontrivial direct product decomposition of the centerless, directly indecomposable group S_k. Thus C, as well as L, is an interval of labels. The one mark in R is adjacent to L. It cannot centralize D, since its membership in the block stabilizer would then put it in B, contradicting independence. It is therefore also adjacent to C. The two intervals and this one intervening mark occupy all the $r - 1$ labels of H consecutively. The omitted label is at an end and centralizes at least one of B, D in full. A permutation centralizing D preserves its orbits, which are the original blocks. A permutation centralizing the diagonal S_m fixes each original block, as in the previous paragraph. Either way G preserves the block system, again a contradiction. Only (2) remains. Its last two conclusions follow from Whiston's equality clause and the two occupied neighbours of L. □

2 Three elementary centralizer arguments

Lemma 3. *Let V be a vector space over $\mathbb{F}_2$ and let m be even. Every nonzero S_m-invariant subspace of V^m, with S_m permuting coordinates, contains a nonzero constant vector.*

Proof. A nonzero constant vector already has the required form. Otherwise take a vector whose i and j coordinates differ. Adding its image under (i, j) gives a vector with the same nonzero value v at i, j and zero elsewhere. Its conjugates give v at any chosen pair of positions. Partition the m positions into pairs and add the corresponding vectors. The result is $(v, \ldots, v)$. □

Lemma 4. *Let $m \geq 4$ be even. Suppose $B \leq A_{3m}$ preserves m blocks of size three and induces S_m on them. Let z be a permutation acting as a transposition in each block. If B centralizes z, then $z \in B$.*

Proof. Choose coordinates in which z is the same transposition on each block. The centralizer of that transposition in S_3 is its own cyclic group. Thus B lies in $C_2^m \rtimes S_m$, in the degree-$3m$ action with one fixed point per block for the base group. Its kernel K on blocks is an S_m-invariant subspace of $\mathbb{F}_2^m$. If $K \neq 0$, Lemma 3 puts its nonzero constant vector z in K, as claimed.

If $K = 0$, the projection identifies B with S_m. Let $t \in B$ be the unique lift of a transposition of two blocks. It is an involution, and on the two exchanged blocks it is a product of three transpositions. The lifts of the permutations of the other $m - 2$ blocks centralize t and act

transitively on those blocks. Consequently its restrictions to the fixed blocks are conjugate. Each restriction is either the identity or the transposition defining z, so all have the same parity. There are an even number $m-2$ of these blocks. Their total contribution is even, whereas the contribution from the exchanged blocks is odd. Thus t is odd, contradicting $B \leq A_{3m}$. □

Lemma 5. *Suppose B acts on sixteen points, preserves four blocks of size four, and maps isomorphically onto S_4 in its block action. If a nontrivial group D fixes the blocks and centralizes B, then the full centralizer of B fixes each block.*

Proof. Identify B with S_4 through the block action and fix one block. Its stabilizer is S_3. A B-orbit meets this block in an orbit of that S_3, of length $d \leq 4$ dividing six. Thus $d = 1, 2, 3$ and the full orbit has length $4d$. The corresponding point stabilizers are S_3, A_3, C_2, respectively.

If a twelve-point orbit occurs, the remaining orbit has four points. On the twelve-point orbit, permutations commuting with B and fixing blocks correspond to $N_{S_3}(C_2)/C_2 = 1$. On the four-point orbit the analogous quotient is $N_{S_3}(S_3)/S_3 = 1$. Unequal orbit sizes prevent any interchange of the two orbits. This would make D trivial, a contradiction.

All orbits therefore have length four or eight. For their point stabilizers $J = S_3, A_3$ we have $N_{S_4}(J) = S_3$, the block stabilizer. In the coset action, a commuting permutation is given by an element of $N_{S_4}(J)/J$ and hence fixes every block. The same calculation applies to equivariant bijections between two isomorphic orbits: choose their base points in the same original block, so both stabilizers are the same $J \leq S_3$. The element defining an equivariant bijection again belongs to $N_{S_4}(J) \leq S_3$. Every permutation centralizing B is assembled from these within-orbit maps and equivariant interchanges. It therefore fixes each block, as asserted. □

Lemma 6. *Suppose B, D preserve four blocks of size four, commute, and have $B \cap D = 1$. Suppose B induces S_4 on the blocks, while D fixes them and has a faithful restriction generated by two independent involutions. Then B acts faithfully on the blocks, and its full centralizer fixes each block.*

Proof. Transport coordinates by B so that D acts identically on all blocks. Two distinct independent involutions in S_4 generate a dihedral group of order four, six or eight, since the order of their product is two, three or four. The order-six group is an S_3 fixing a point, and has trivial centralizer in S_4. The order-eight group is a Sylow subgroup, whose centralizer is its central involution. The order-four groups are either the regular Klein group or the group generated by two disjoint transpositions; their centralizer is the group itself. These elementary centralizers can be obtained by preserving the orbits and commuting with the two generators. In every case $Z = C_{S_4}(D)$ is an elementary abelian two-group contained in D, possibly trivial.

Now $B \leq Z^4 \rtimes S_4$. Since Z is abelian, conjugation by B on its block kernel $K \leq Z^4$ is the ordinary permutation of coordinates. If $K \neq 0$, Lemma 3 supplies a nontrivial diagonal element of Z in K. That element belongs to both B and D, a contradiction. Thus $K = 0$. Lemma 5 now applies. □

3 Small even actions

Lemma 7. *Three independent involutions in A_5 generate A_5.*

Proof. An intransitive subgroup is contained in A_4 or in the even part of $S_3 \times S_2$, which is S_3. In A_4 all involutions lie in its Klein subgroup; neither this group nor S_3 has three independent involutions. A proper transitive subgroup has order divisible by five and dividing sixty, hence order $5, 10, 15, 20$ or 30. Orders twenty and thirty would give faithful nontrivial actions of the simple group A_5 on three or two cosets, which is impossible. A group of order fifteen has normal Sylow three- and five-subgroups and hence an element of order fifteen, absent from A_5. Order five supplies no involutions; at order ten, two distinct involutions generate the whole dihedral group. The only remaining possibility is A_5 itself. □

Lemma 8. *No subgroup of A_6 has a string C-marking of rank four.*

Proof. The transitive case is precisely the acquired L047 finite conclusion specified above. For an intransitive group choose a nonempty invariant set of size $a \leq 3$, passing to its complement if necessary. It is contained in $(S_a \times S_{6-a}) \cap A_6$. For $a = 1$ this is A_5, whose independent length is at most three by Whiston's equality clause in S_5. For $a = 2$ it is isomorphic to S_4, since the permutation on the two-point factor is determined by the sign on the other factor; Whiston again bounds the independent length by three. For $a = 3$ its order is eighteen, so every strictly increasing subgroup chain has at most three steps. An independent family produces such a chain and has at most three members. All cases exclude rank four. □

4 Proof of the rank bound

Proof of Theorem 1. At $n = 12$ the complete acquired A_{12} census gives $r \leq 5$. Assume $n > 12$. The disconnected case was settled before (1). If (1) does not already imply $r \leq \lfloor (km - 1)/2 \rfloor$, then

$$(k, m) \in \{(3, 5), (3, 6), (4, 4), (5, 3), (6, 3)\}.$$

Indeed, $2(k + m) \geq km$ is necessary when km is even; when it is odd the failed integer bound requires $2(k+m) \geq km+1$. In particular $(k-2)(m-2) \leq 4$, and listing integers $k, m > 2$ with $km > 12$ gives the five pairs above. In each case failure forces $r = k + m$. Lemma 2 applies, so $P = S_m$, $|C| = k - 2$, $|R| = 2$, and ρ_i centralizes B.

For $(k, m) = (3, 5)$, D has one involutory generator. Its faithful restriction to a block is a transposition. It acts as such on all five blocks and is therefore odd, a contradiction.

For $(3, 6)$, write that generator as z. Lemma 4 puts z in B, contrary to $B \cap D = 1$.

For $(5, 3)$, the three independent involutions of D restrict to even permutations on a block, since the number of identical blocks is odd. Lemma 7 gives $D = A_5$ diagonally. Its centralizer in S_5 is trivial, so B is the diagonal faithful S_3 on the three blocks. A block transposition is a product of five transpositions on points and is odd. Again this contradicts $B \leq A_{15}$.

For $(6, 3)$, the four generators of D similarly restrict to a string C-marking of rank four in A_6. Restriction is faithful and preserves the intersection property. Lemma 8 excludes it.

Finally, for $(4, 4)$, D has two independent involutory generators. Lemma 6 says that every permutation centralizing B fixes each original block. In particular ρ_i does so. Together with H, it generates a group preserving that nontrivial block system, contradicting $G = A_{16}$. This proves the theorem in every case. □

Scope and attribution. The source's rank conclusion is recovered only when both block parameters exceed two. The cases of two-point blocks and of two blocks remain separate obligations. This component uses no induction on alternating degree and no new finite search. The L047 decomposition, Whiston input, classical simplicity of A_5, and acquired A_6, A_{12} classifications remain explicitly borrowed. The equality and centralizer arguments above are this continuation; no priority claim is made for their elementary ingredients. No whole-original, open-question, Lean or numerical compression credit is assigned by this component.